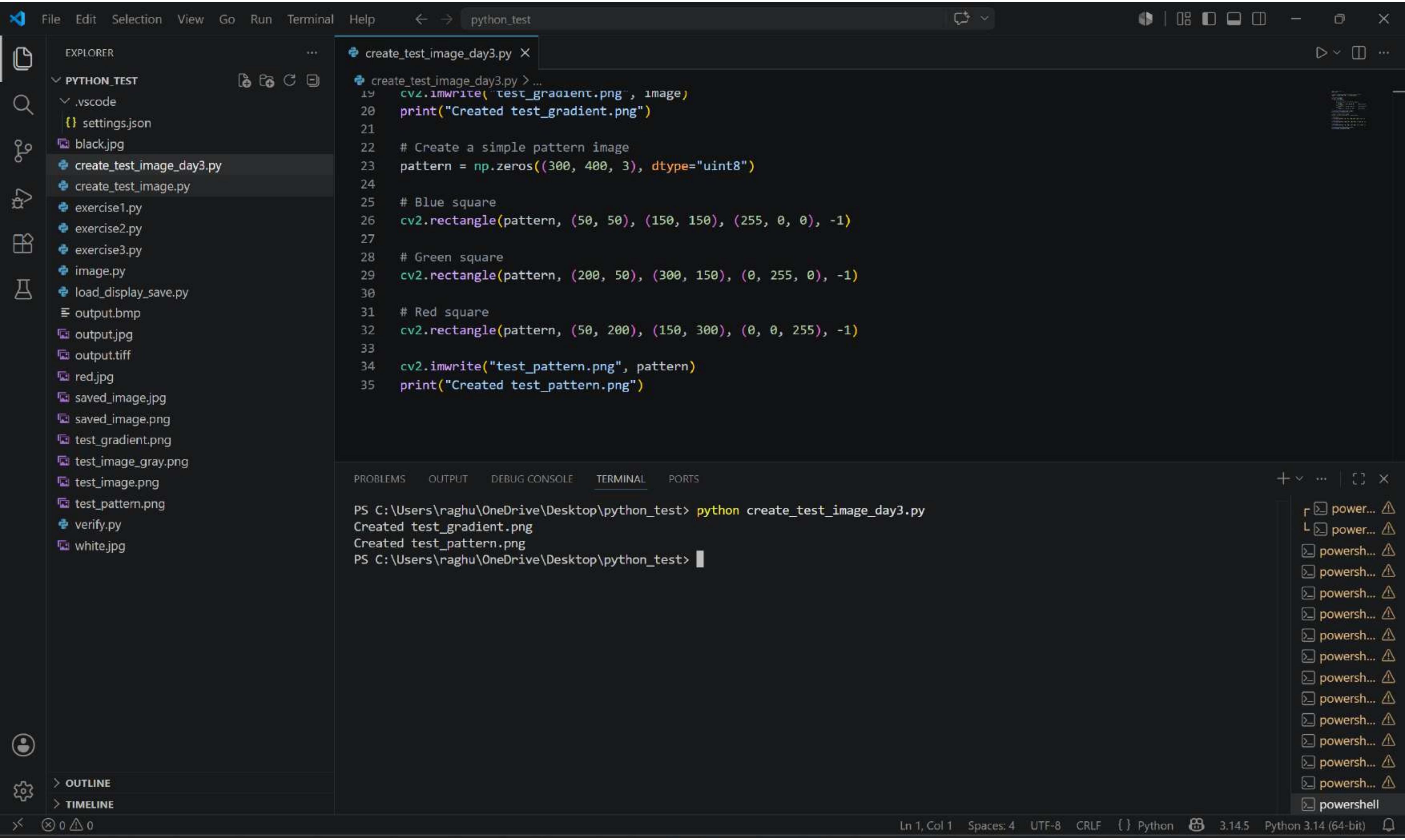
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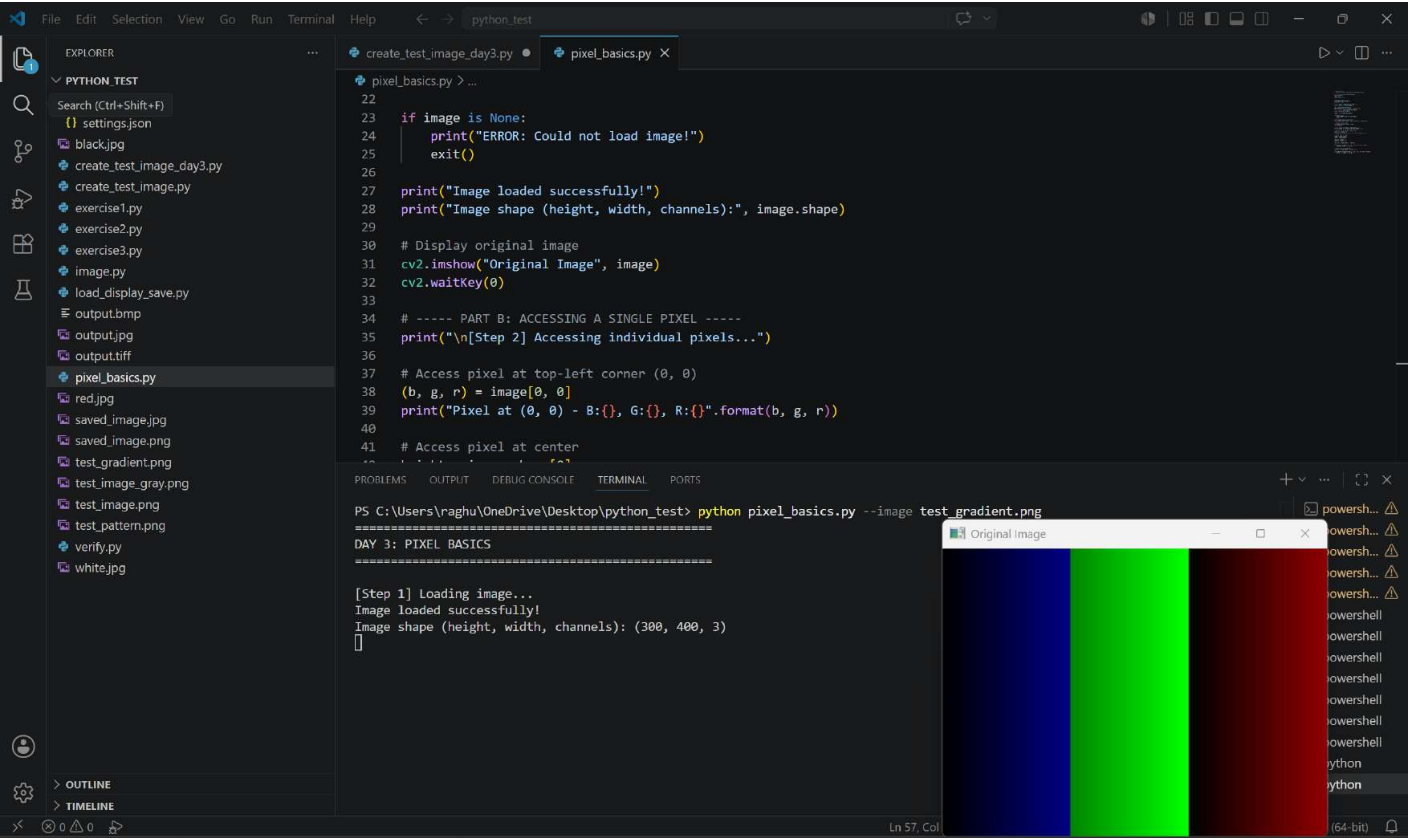
OUTLINE
TIMELINE
create_test_image_day3.py
create_test_image_day3.py > ...
cv2.imwrite("test_gradient.png", image)
print("Created test_gradient.png")
# Create a simple pattern image
pattern = np.zeros((300, 400, 3), dtype="uint8")
# Blue square
cv2.rectangle(pattern, (50, 50), (150, 150), (255, 0, 0), -1)
# Green square
cv2.rectangle(pattern, (200, 50), (300, 150), (0, 255, 0), -1)
# Red square
cv2.rectangle(pattern, (50, 200), (150, 300), (0, 0, 255), -1)
cv2.imwrite("test_pattern.png", pattern)
print("Created test_pattern.png")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python create_test_image_day3.py
Created test_gradient.png
Created test_pattern.png
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)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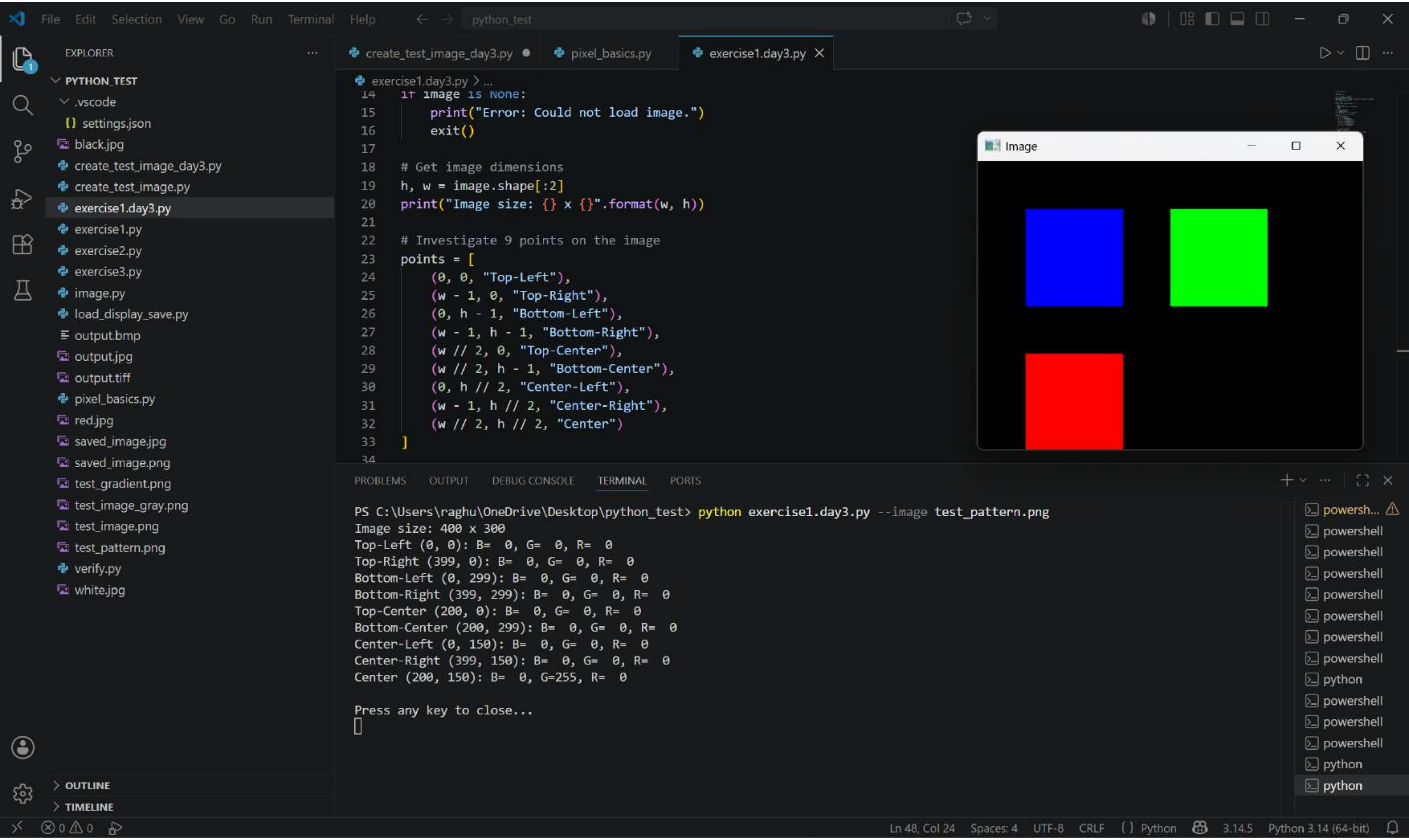
OUTLINE
TIMELINE
create_test_image_day3.py
pixel_basics.py
pixel_basics.py > ...
if image is None:
    print("ERROR: Could not load image!")
    exit()
print("Image loaded successfully!")
print("Image shape (height, width, channels):", image.shape)
# Display original image
cv2.imshow("Original Image", image)
cv2.waitKey(0)
# ----- PART B: ACCESSING A SINGLE PIXEL -----
print("\n[Step 2] Accessing individual pixels...")
# Access pixel at top-left corner (0, 0)
(b, g, r) = image[0, 0]
print("Pixel at (0, 0) - B:{}, G:{}, R:{}".format(b, g, r))
# Access pixel at center
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python pixel_basics.py --image test_gradient.png
==
DAY 3: PIXEL BASICS
==
[Step 1] Loading image...
Image loaded successfully!
Image shape (height, width, channels): (300, 400, 3)
Original Image
Ln 57, Col
(64-bit)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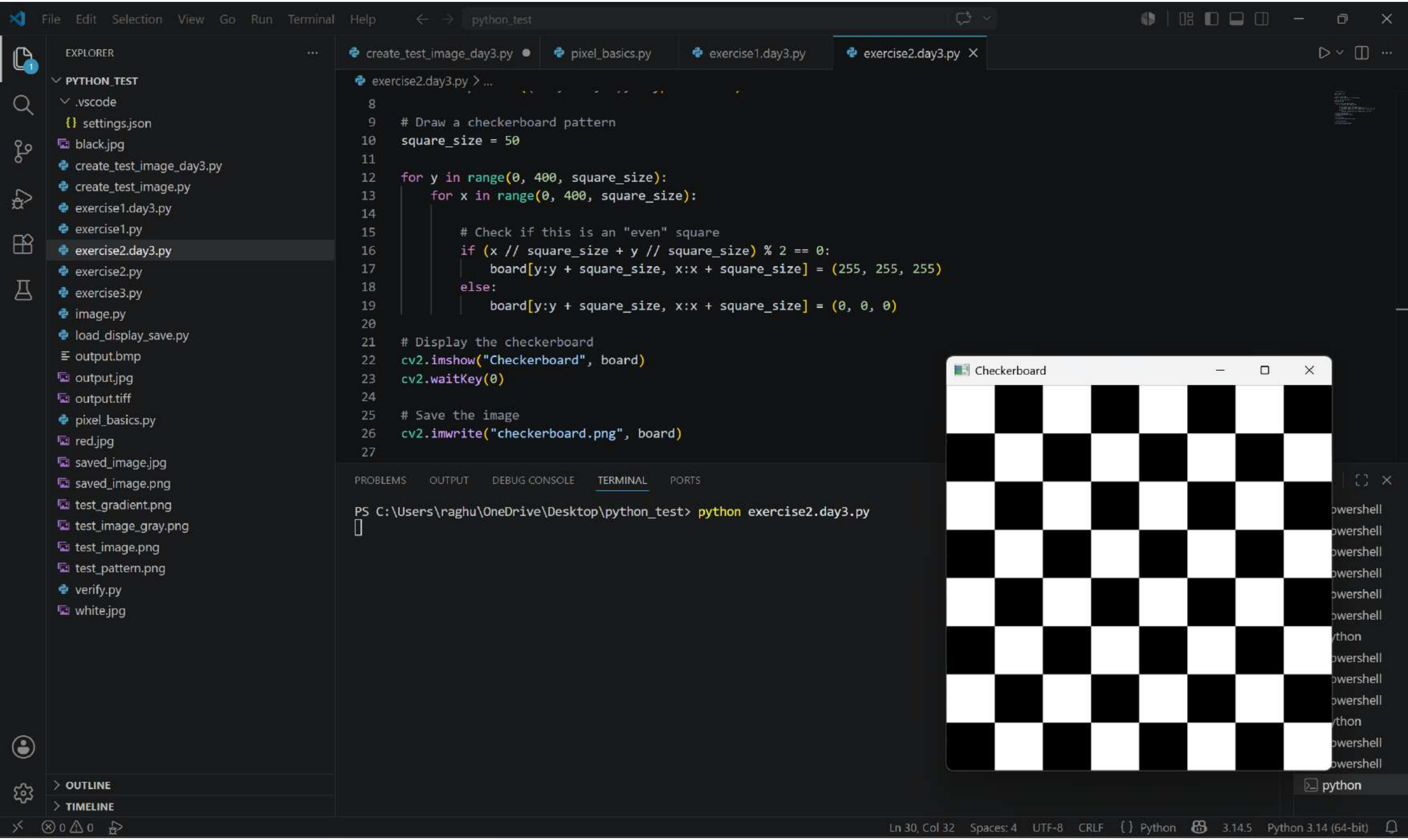
OUTLINE
TIMELINE
exercise1.day3.py > ...
if image is None:
print("Error: Could not load image.")
exit()
# Get image dimensions
h, w = image.shape[:2]
print("Image size: {} x {}".format(w, h))
# Investigate 9 points on the image
points = [
(0, 0, "Top-Left"),
(w - 1, 0, "Top-Right"),
(0, h - 1, "Bottom-Left"),
(w - 1, h - 1, "Bottom-Right"),
(w // 2, 0, "Top-Center"),
(w // 2, h - 1, "Bottom-Center"),
(0, h // 2, "Center-Left"),
(w - 1, h // 2, "Center-Right"),
(w // 2, h // 2, "Center")
]
Image
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise1.day3.py --image test_pattern.png
Image size: 400 x 300
Top-Left (0, 0): B= 0, G= 0, R= 0
Top-Right (399, 0): B= 0, G= 0, R= 0
Bottom-Left (0, 299): B= 0, G= 0, R= 0
Bottom-Right (399, 299): B= 0, G= 0, R= 0
Top-Center (200, 0): B= 0, G= 0, R= 0
Bottom-Center (200, 299): B= 0, G= 0, R= 0
Center-Left (0, 150): B= 0, G= 0, R= 0
Center-Right (399, 150): B= 0, G= 0, R= 0
Center (200, 150): B= 0, G=255, R= 0
Press any key to close...
powershell
python
Ln 48, Col 24 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)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OUTLINE
TIMELINE
create_test_image_day3.py
pixel_basics.py
exercise1.day3.py
exercise2.day3.py
exercise2.day3.py > ...
# Draw a checkerboard pattern
square_size = 50
for y in range(0, 400, square_size):
for x in range(0, 400, square_size):
# Check if this is an "even" square
if (x // square_size + y // square_size) % 2 == 0:
board[y:y + square_size, x:x + square_size] = (255, 255, 255)
else:
board[y:y + square_size, x:x + square_size] = (0, 0, 0)
# Display the checkerboard
cv2.imshow("Checkerboard", board)
cv2.waitKey(0)
# Save the image
cv2.imwrite("checkerboard.png", board)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise2.day3.py
Checkerboard
python
Ln 30, Col 32
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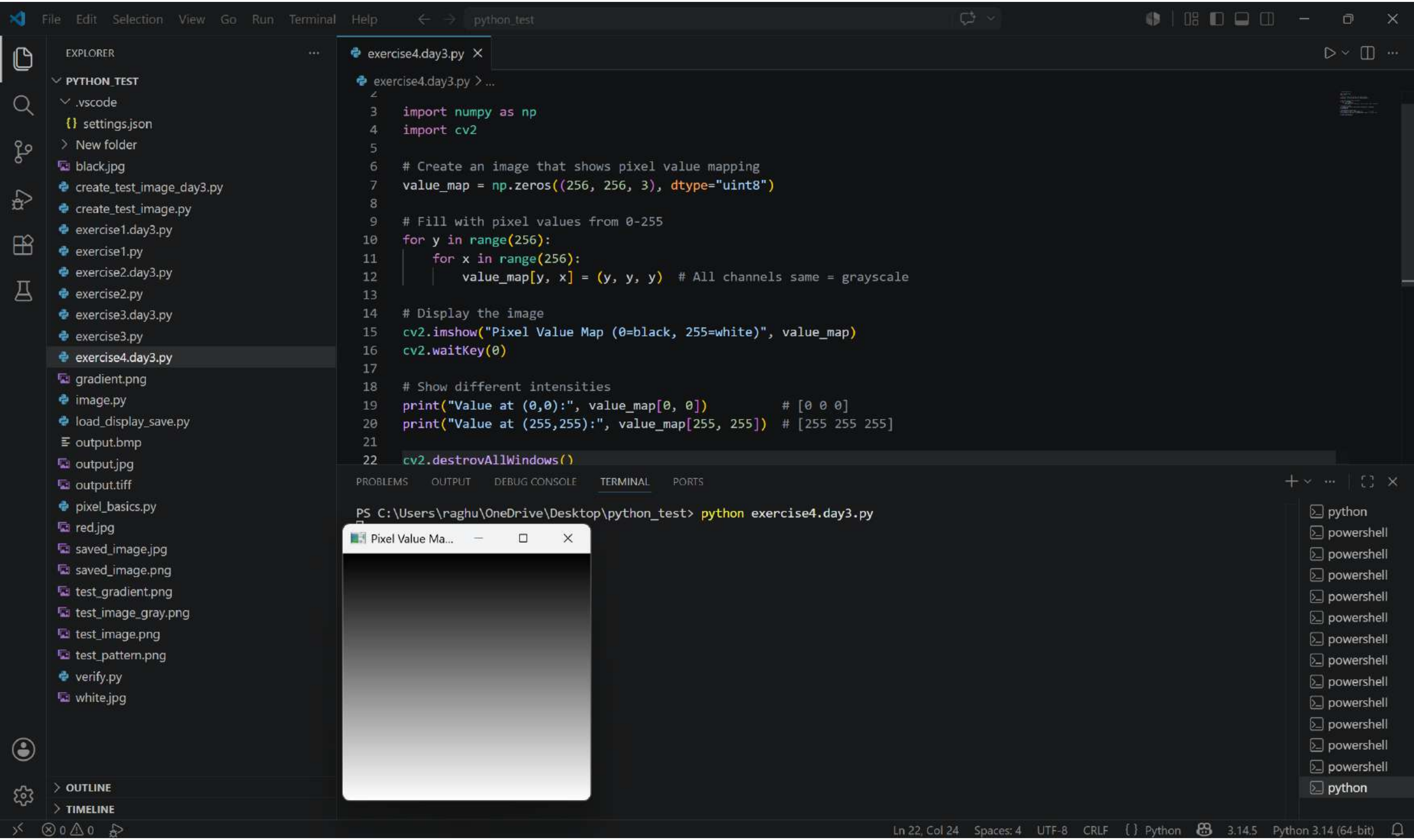
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OUTLINE
TIMELINE
exercise4.day3.py
exercise4.day3.py > ...
import numpy as np
import cv2
# Create an image that shows pixel value mapping
value_map = np.zeros((256, 256, 3), dtype="uint8")
# Fill with pixel values from 0-255
for y in range(256):
for x in range(256):
value_map[y, x] = (y, y, y) # All channels same = grayscale
# Display the image
cv2.imshow("Pixel Value Map (0=black, 255=white)", value_map)
cv2.waitKey(0)
# Show different intensities
print("Value at (0,0):", value_map[0, 0])   # [0 0 0]
print("Value at (255,255):", value_map[255, 255]) # [255 255 255]
cv2.destroyAllWindows()
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise4.day3.py
Pixel Value Ma...
python
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
python
Ln 22, Col 24
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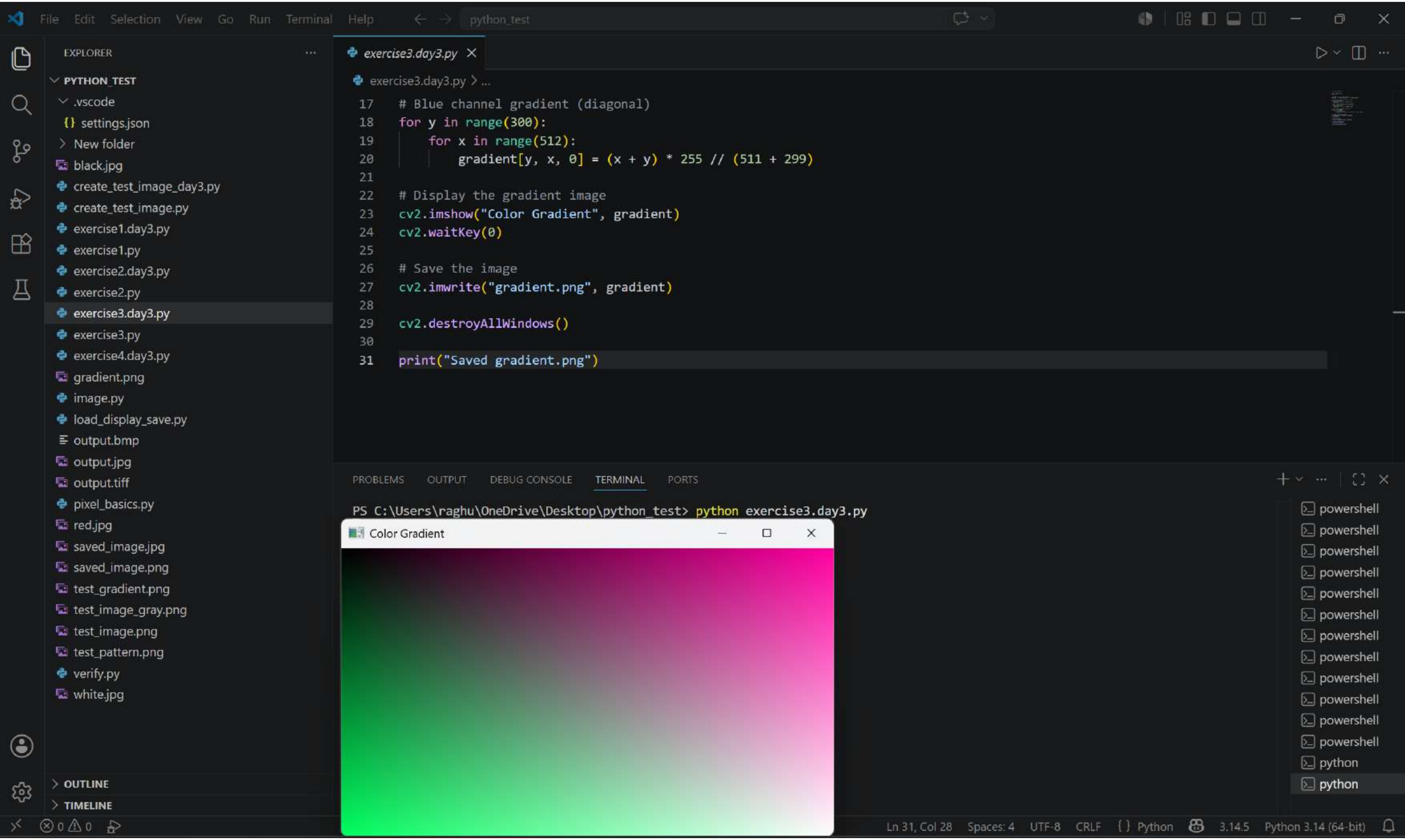
Python
3.14.5
Python 3.14 (64-bit)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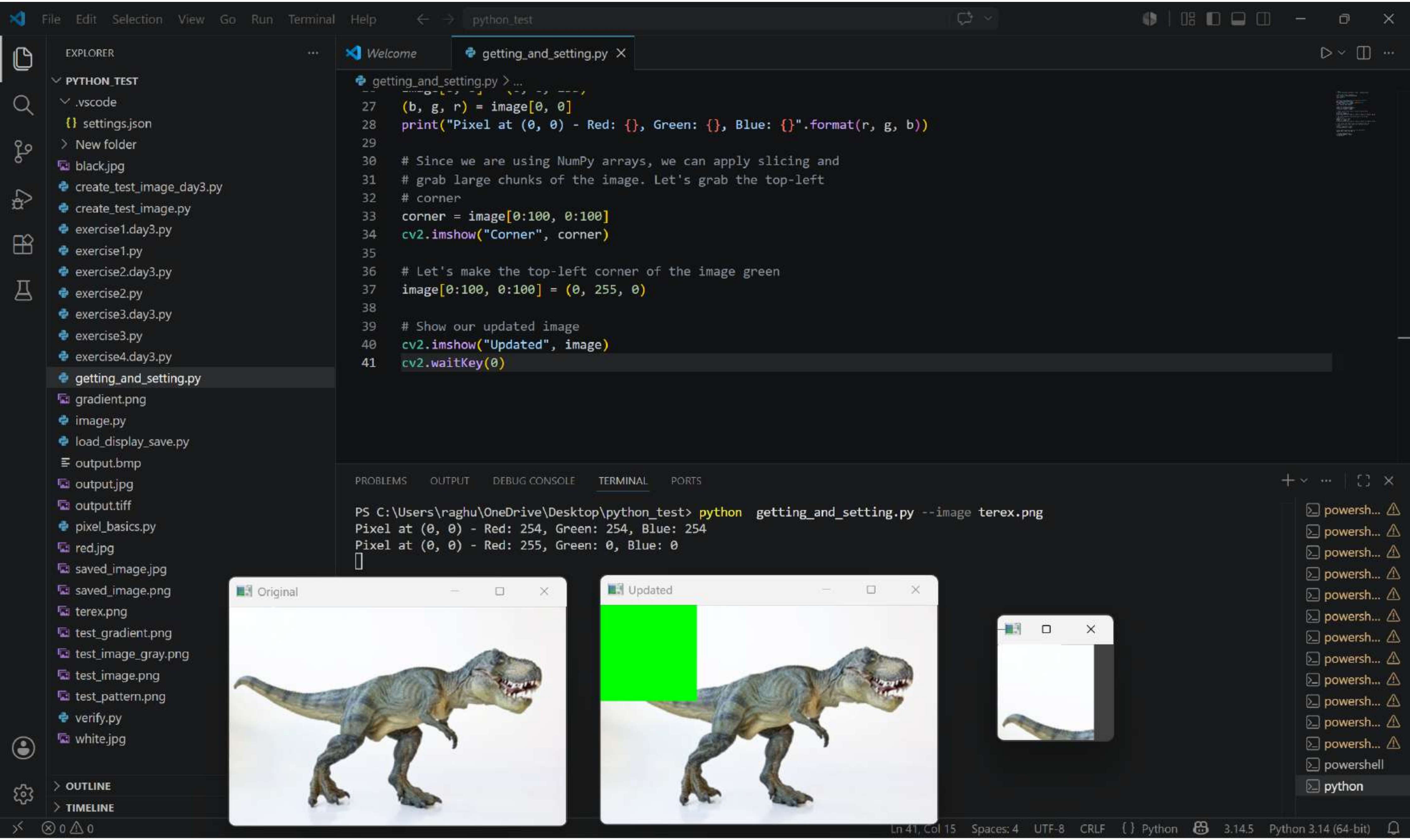
OUTLINE
TIMELINE
exercise3.day3.py
exercise3.day3.py > ...
# Blue channel gradient (diagonal)
for y in range(300):
for x in range(512):
gradient[y, x, 0] = (x + y) * 255 // (511 + 299)
# Display the gradient image
cv2.imshow("Color Gradient", gradient)
cv2.waitKey(0)
# Save the image
cv2.imwrite("gradient.png", gradient)
cv2.destroyAllWindows()
print("Saved gradient.png")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise3.day3.py
Color Gradient
powershell
python
Ln 31, Col 28 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)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OUTLINE
TIMELINE
Welcome
getting_and_setting.py
(b, g, r) = image[0, 0]
print("Pixel at (0, 0) - Red: {}, Green: {}, Blue: {}".format(r, g, b))
# Since we are using NumPy arrays, we can apply slicing and
# grab large chunks of the image. Let's grab the top-left
# corner
corner = image[0:100, 0:100]
cv2.imshow("Corner", corner)
# Let's make the top-left corner of the image green
image[0:100, 0:100] = (0, 255, 0)
# Show our updated image
cv2.imshow("Updated", image)
cv2.waitKey(0)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python getting_and_setting.py --image terex.png
Pixel at (0, 0) - Red: 254, Green: 254, Blue: 254
Pixel at (0, 0) - Red: 255, Green: 0, Blue: 0
Original
Updated
powershell
python
Ln 41, Col 15 Spaces: 4 UTF-8 CRLF {} Python 3.14.5 Python 3.14 (64-bit)